

❖ EBS Volume Management and Backup Strategies

❖ What is Amazon EBS?

- Amazon **EBS** (Elastic Block Store) is a **block-level storage service** provided by **AWS** that is used to store persistent data for Amazon EC2 (Elastic Compute Cloud) instances.
- EBS volumes behave like traditional **hard drives** or **SSDs** attached to **virtual machines**, and they persist independently of the EC2 instance's lifecycle.
- EBS is designed to provide high-performance, scalable, and durable storage that can be attached to a single EC2 instance at a time (though multi-attach is possible with certain volume types).
- It supports both **boot volumes** and **data volumes**, and you can take **snapshots** for backup, recovery, and replication.

▪ EBS Key Features

1. Persistent Storage

Data remains available even after the EC2 instance is stopped or terminated.

2. Performance-Optimized

Offers SSD and HDD-based storage options tailored for transactional or throughput-intensive workloads.

3. Scalable

EBS volumes can be resized and their performance can be adjusted without downtime.

4. Snapshots and Backup

EBS allows point-in-time snapshots that are stored in Amazon S3 and can be used to restore volumes or create new ones.

5. Encryption

Supports encryption at rest and in transit using AWS Key Management Service (KMS).

6. Availability and Durability

EBS volumes are automatically replicated within the same Availability Zone to prevent data loss due to hardware failure.

7. Attach and Detach Volumes

Volumes can be attached to or detached from instances as needed.

8. Integration with Other AWS Services

Works seamlessly with EC2, CloudWatch, IAM, Auto Scaling, and more.

❖ Types of EBS Volumes

▪ Volume types

1. Solid state drive (SSD) volumes
2. Hard disk drive (HDD) volumes
3. Previous generation volumes

Uses of Each volume type see on Day 6 EBS Volume

https://edublitz.edmingle.com/v3/myaccount/course/146217/lesson/1479706?lesson=1479706§ion=351121&class_id=352537

❖ What is the difference between EBS Snapshots and AMIs?

- **EBS Snapshot:** A backup of a specific EBS volume that can be used to **restore** or **create new volumes**.
- **AMI (Amazon Machine Image):** A **pre-configured template** that **includes** an operating system, application software, and EBS snapshots, allowing instances to be launched with **predefined** settings.

❖ Can you attach an EBS volume to multiple EC2 instances at the same time? If yes, how?

- Yes, but only with **EBS Multi-Attach** feature, which is supported for **Provisioned IOPS SSD (io2)** volumes.
- **Multiple EC2 instances** within the **same Availability Zone (AZ)** can simultaneously access a single io2 volume.
- Applications must be designed to **handle concurrent access** (e.g. clustered databases or shared file systems).

❖ How does EBS handle data consistency for snapshots of running instances?

- When you take a **snapshot** of an EBS volume attached to a **running EC2 instance**, Amazon EBS ensures **crash-consistent** snapshots — but not **application-consistent**, unless you take additional steps.

Example: **1. Crash-Consistent Snapshots (Default Behavior)**

- EBS snapshots **capture the volume as if the power was suddenly cut**.
- All written data is already saved to disk

❖ What happens when you delete an EBS volume? Can you recover the data?

Answer:

- If an EBS volume is **deleted** without snapshots, the data is **permanently lost**.
- If snapshots **exist**, a new volume can be **created** from the latest snapshot to **restore** the data.
- AWS **does not provide** soft delete or recycle bin for EBS volumes unless snapshots are taken.

❖ What is the difference between AWS EBS gp2 and gp3 volumes?

Answer:

- **gp2 (General Purpose SSD)**: Performance scales with **volume size**(3 IOPS/GB with bursting).
- **gp3 (Next-Generation SSD)**: Performance (IOPS and throughput) is **decoupled** from **storage size**, allowing manual provisioning of **IOPS** and throughput for **better cost** optimization.

❖ What is the recommended method to migrate an EBS volume from one region to another?

Answer:

- **Create** an **EBS Snapshot** of the volume.
- **Copy** the snapshot to the **target region** using the AWS Console or CLI.
- **Create** a new EBS volume from the **copied** snapshot in the **target region**.

❖ What happens when an EBS volume reaches its maximum IOPS or throughput limit?

Answer:

- When the limits are reached, **disk I/O operations experience increased** latency or are throttled.
- To mitigate this:
 - Upgrade to **Provisioned IOPS SSD (io1/io2)** for higher throughput.
 - Optimize application I/O patterns to **reduce** excessive read/write operations.
 - Enable **Elastic Volumes** to dynamically increase volume size and IOPS.

❖ How do you prevent data loss in EC2 instances?

Answer:

Regular EBS Snapshots for backup.

- Use Auto Recovery to **restart** instances on failure.
- Cross-region **replication** for high availability.
- Instance Store-backed instances should not be used for critical data, as they are **non-persistent**.